Time-convolutionless Projection Operator

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1 Nakajima-Zwanzig Equation

Consider an open system S coupled to a bath B. The Hamiltonian for the system, bath, and system–bath coupling are denoted as $\hat{H}_{\mathrm{S}}$, $\hat{H}_{\mathrm{B}}$, and $\hat{H}_{\mathrm{I}}$, respectively. Let $\hat{H}_0 = \hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{B}}$, and introducing a dimensionless parameter α, which will be used later for perturbative expansion, the total Hamiltonian of the composite system is

$$\hat{H} = \hat{H}_0 + \alpha \hat{H}_{\mathrm{I}} \tag{1}$$

and the equation of motion of the density operator for the total system $\hat{\rho}(t)$ in the interaction picture is

$$\frac{\partial}{\partial t}\hat{\rho}(t) = -i\alpha[\hat{H}_{\mathrm{I}}(t), \hat{\rho}(t)] \equiv \alpha\mathcal{L}(t)\hat{\rho}(t) \tag{2}$$

where $\hat{H}_{\mathrm{I}}(t) = e^{i\hat{H}_0 t}\hat{H}_{\mathrm{I}}e^{-i\hat{H}_0 t}$, $\mathcal{L}(t) = -i[\hat{H}_{\mathrm{I}}(t), \bullet]$ is the Liouville superoperator. We have set $\hbar = 1$ in these expressions.

Let's define the projection superoperators $\mathcal{P}$ and $\mathcal{Q} = 1 - \mathcal{P}$, such that for an arbitrary operator $\hat{O}$

$$\boxed{\mathcal{P}\hat{O} = \mathrm{Tr}_{\mathrm{B}}(\hat{O}) \otimes \hat{\rho}_{\mathrm{B}}} \tag{3}$$

where $\hat{\rho}_{\mathrm{B}}$ is some fixed density operator for the bath. Specifically,

$$\mathcal{P}\hat{\rho}(t) = \mathrm{Tr}_{\mathrm{B}}[\hat{\rho}(t)] \otimes \hat{\rho}_{\mathrm{B}} \equiv \hat{\rho}_{\mathrm{S}}(t) \otimes \hat{\rho}_{\mathrm{B}} \tag{4}$$

$\hat{\rho}_{\mathrm{S}}(t)$ is the reduced density operator for the system. Applying the projection superoperators to Eq. (2), and inserting the identity $1 = \mathcal{P} + \mathcal{Q}$ between the Liouvile superoperator and the $\hat{\rho}(t)$, we have

$$\frac{\partial}{\partial t}\mathcal{P}\hat{\rho}(t) = \alpha\mathcal{P}\mathcal{L}(t)\mathcal{P}\hat{\rho}(t) + \alpha\mathcal{P}\mathcal{L}(t)\mathcal{Q}\hat{\rho}(t) \tag{5a}$$

$$\frac{\partial}{\partial t}\mathcal{Q}\hat{\rho}(t) = \alpha\mathcal{Q}\mathcal{L}(t)\mathcal{P}\hat{\rho}(t) + \alpha\mathcal{Q}\mathcal{L}(t)\mathcal{Q}\hat{\rho}(t) \tag{5b}$$

Once we obtain a formal solution for $\mathcal{Q}\hat{\rho}(t)$ from Eq. (5b), then substituting the obtained expression into Eq. (5a) gives a closed equation for $\mathcal{P}\hat{\rho}(t)$.

Eq. (5b) is a first-order inhomogeneous differential equation for $\mathcal{Q}\hat{\rho}(t)$. It's corresponding homogeneous differential equation is

$$\frac{\partial}{\partial t}\mathcal{Q}\hat{\rho}(t) = \alpha\mathcal{Q}\mathcal{L}(t)\mathcal{Q}\hat{\rho}(t) \tag{6}$$

The formal solution for this homogeneous differential equation is

$$\mathcal{Q}\hat{\rho}(t)|_{\text{homogeneous}} = \mathcal{G}(t,t_0)\mathcal{Q}\hat{\rho}(t_0) \tag{7}$$

where

$$\boxed{\mathcal{G}(t,t') = \hat{T}_{\leftarrow} \exp\left[\alpha \int_{t'}^{t} \mathrm{d}\tau\, \mathcal{Q}\mathcal{L}(\tau)\right]} \tag{8}$$

is the propagator. **The operator $\hat{T}_{\leftarrow}$ ensures a chronological time order, where the time argument in the expression increases from right to left.** The propagator satisfies the following differential equation

$$\frac{\partial}{\partial t}\mathcal{G}(t,t') = \alpha\mathcal{Q}\mathcal{L}(t)\mathcal{G}(t,t') \tag{9}$$

The propagator also has the properties $\mathcal{G}(t,t) = 1$ and $\mathcal{G}(t_3,t_1) = \mathcal{G}(t_3,t_2)\mathcal{G}(t_2,t_1)$ for $t_3 > t_2 > t_1$.

Using the general procedure of solving first-order inhomogeneous differential equations, we suppose the solution for Eq. (5b) has the form

$$\mathcal{Q}\hat{\rho}(t) = \mathcal{G}(t,t_0)c(t) \tag{10}$$

where $c(t)$ is an unknown function and $c(t_0) = \mathcal{Q}\hat{\rho}(t_0)$. Substituting this test solution into Eq. (5b), we get the differential equation for $c(t)$,

$$\mathcal{G}(t,t_0)\frac{\partial}{\partial t}c(t) = \alpha\mathcal{Q}\mathcal{L}(t)\mathcal{P}\hat{\rho}(t) \tag{11}$$

The solution of $c(t)$ is

$$c(t) = \alpha \int_{t_0}^{t} \mathrm{d}t'\, \mathcal{G}^{-1}(t',t_0)\mathcal{Q}\mathcal{L}(t')\mathcal{P}\hat{\rho}(t') + \mathcal{Q}\hat{\rho}(t_0) \tag{12}$$

The final solution for Eq. (5b) is thus

$$\mathcal{Q}\hat{\rho}(t) = \mathcal{G}(t,t_0)\mathcal{Q}\hat{\rho}(t_0) + \alpha \int_{t_0}^{t} \mathrm{d}t'\, \mathcal{G}(t,t')\mathcal{Q}\mathcal{L}(t')\mathcal{P}\hat{\rho}(t') \tag{13}$$

We have used that $\mathcal{G}(t,t_0)\mathcal{G}^{-1}(t',t_0) = \mathcal{G}(t,t')$ for $t > t'$.

Finally, substituting the formal solution Eq. (13) into Eq. (5a), we obtain the **Nakajima-Zwanzig equation**,

$$\boxed{\frac{\partial}{\partial t}\mathcal{P}\hat{\rho}(t) = \alpha\mathcal{P}\mathcal{L}(t)\mathcal{P}\hat{\rho}(t) + \alpha\mathcal{P}\mathcal{L}(t)\mathcal{G}(t,t_0)\mathcal{Q}\hat{\rho}(t_0) + \alpha^2 \int_{t_0}^{t} \mathrm{d}t'\, \mathcal{P}\mathcal{L}(t)\mathcal{G}(t,t')\mathcal{Q}\mathcal{L}(t')\mathcal{P}\hat{\rho}(t')} \tag{14}$$

This is a integro-differential equation for $\mathcal{P}\hat{\rho}(t)$. The second term, $\alpha\mathcal{P}\mathcal{L}(t)\mathcal{G}(t,t_0)\mathcal{Q}\hat{\rho}(t_0)$ is the inhomogeneous term, as it depends on t but is independent of $\mathcal{P}\hat{\rho}(t)$.

In many cases, we can **assume that the odd moments of the system–bath coupling Hamiltonian with respect to $\hat{\rho}_{\mathrm{B}}$ vanish**,

$$\mathrm{Tr}_{\mathrm{B}}\left(\hat{H}_{\mathrm{I}}(t_1)\hat{H}_{\mathrm{I}}(t_2)\cdots\hat{H}_{\mathrm{I}}(t_{2n+1})\hat{\rho}_{\mathrm{B}}\right) = 0, \quad \forall\, t_1, t_2, \cdots, t_{2n+1} \tag{15}$$

which lead to the relation

$$\boxed{\mathcal{P}\mathcal{L}(t_1)\mathcal{L}(t_2)\cdots\mathcal{L}(t_{2n+1})\mathcal{P} = 0, \quad \forall\, t_1, t_2, \cdots, t_{2n+1}} \tag{16}$$

This relation can simplify the perturbative expansion of Eq. (14). For example, the first term, which involves $\mathcal{P}\mathcal{L}(t)\mathcal{P}$, vanishes.

For a factorizing initial contition, $\hat{\rho}(t_0) = \hat{\rho}_{\mathrm{S}}(t_0) \otimes \hat{\rho}_{\mathrm{B}}$**, we have** $\mathcal{P}\hat{\rho}(t_0) = \hat{\rho}(t_0)$ **and** $\mathcal{Q}\hat{\rho}(t_0) = 0$**. In this case, the inhomogeneous term in Eq. (14) becomes zero.** The expansion of $\mathcal{G}(t, t')$ to the first order of α is

$$\mathcal{G}(t, t') \approx 1 + \alpha \int_{t'}^{t} \mathrm{d}\tau\, \mathcal{Q}\mathcal{L}(\tau) \tag{17}$$

Then

$$\alpha^2 \mathcal{P}\mathcal{L}(t)\mathcal{G}(t, t')\mathcal{Q}\mathcal{L}(t')\mathcal{P} \approx \alpha^2 \mathcal{P}\mathcal{L}(t)\mathcal{Q}\mathcal{L}(t')\mathcal{P} + O(\alpha^3) \tag{18}$$

When Eq. (16) is valid, we have $\mathcal{P}\mathcal{L}(t) = \mathcal{P}\mathcal{L}(t)(\mathcal{P} + \mathcal{Q}) = \mathcal{P}\mathcal{L}(t)\mathcal{Q}$. **The equation of motion of second order for** $\mathcal{P}\hat{\rho}(t)$ is thus

$$\boxed{\frac{\partial}{\partial t}\mathcal{P}\hat{\rho}(t) = \alpha^2 \int_{t_0}^{t} \mathrm{d}t'\, \mathcal{P}\mathcal{L}(t)\mathcal{L}(t')\mathcal{P}\hat{\rho}(t')} \tag{19}$$

or explicitly in terms of the commutators

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}_{\mathrm{S}}(t) = -\alpha^2 \int_{t_0}^{t} \mathrm{d}t'\, \mathrm{Tr}_{\mathrm{B}}\left([\hat{H}_{\mathrm{I}}(t), [\hat{H}_{\mathrm{I}}(t'), \hat{\rho}_{\mathrm{S}}(t') \otimes \hat{\rho}_{\mathrm{B}}]]\right)} \tag{20}$$

This equation is still an intergo-differential equation. The time convolution in this equation is difficult to treat.

2 Time-convolutionless Projection Operator

The time-convolutionless projection operator method can remove the time convolution integral in the Nakajima-Zwanzig equation. We define the **backward propagator for the total system**,

$$\boxed{G(t, t') = \hat{T}_{\rightarrow} \exp\left[-\alpha \int_{t'}^{t} \mathrm{d}\tau\, \mathcal{L}(\tau)\right]} \tag{21}$$

where $\hat{T}_{\rightarrow}$ **ensures antichronological time order, where the time argument in the expression increases from left to right.** Then the total density operator at time t' can be expressed in terms of the total density operator at time t via

$$\hat{\rho}(t') = G(t, t')(\mathcal{P} + \mathcal{Q})\hat{\rho}(t) \tag{22}$$

Substituting this expression into Eq. (13), we have

$$\mathcal{Q}\hat{\rho}(t) = \mathcal{G}(t, t_0)\mathcal{Q}\hat{\rho}(t_0) + \alpha \int_{t_0}^{t} \mathrm{d}t'\, \mathcal{G}(t, t')\mathcal{Q}\mathcal{L}(t')\mathcal{P}G(t, t')(\mathcal{P} + \mathcal{Q})\hat{\rho}(t) \tag{23}$$

Further defining the superoperator

$$\boxed{\Sigma(t)=\alpha\int_{t_0}^{t}\mathrm{d}t'\,\mathcal{G}(t,t')\mathcal{Q}\mathcal{L}(t')\mathcal{P}G(t,t')} \tag{24}$$

then Eq. (13) becomes

$$[1-\Sigma(t)]\mathcal{Q}\hat{\rho}(t)=\mathcal{G}(t,t_0)\mathcal{Q}\hat{\rho}(t_0)+\Sigma(t)\mathcal{P}\hat{\rho}(t) \tag{25}$$

$\Sigma(t)$ has the following obvious properties: when $t=t_0$, $\Sigma(t)=0$; when $\alpha=0$, $\Sigma(t)=0$. Therefore, **for weak system–bath coupling, or for small $t-t_0$, $\Sigma(t)\ll 1$ and $1-\Sigma(t)$ is invertible.** Thus

$$\mathcal{Q}\hat{\rho}(t)=[1-\Sigma(t)]^{-1}\Sigma(t)\mathcal{P}\hat{\rho}(t)+[1-\Sigma(t)]^{-1}\mathcal{G}(t,t_0)\mathcal{Q}\hat{\rho}(t_0) \tag{26}$$

Substituting this expression into Eq. (5a), and noting that $\alpha\mathcal{P}\mathcal{L}(t)+\alpha\mathcal{P}\mathcal{L}(t)[1-\Sigma(t)]^{-1}\Sigma(t)=\alpha\mathcal{P}\mathcal{L}(t)[1-\Sigma(t)]^{-1}$, we obtain the exact **time-convolutionless (TCL) form of the master equation**

$$\boxed{\frac{\partial}{\partial t}\mathcal{P}\hat{\rho}(t)=\mathcal{K}(t)\mathcal{P}\hat{\rho}(t)+\mathcal{I}(t)\mathcal{Q}\hat{\rho}(t_0)} \tag{27}$$

where

$$\boxed{\mathcal{K}(t)=\alpha\mathcal{P}\mathcal{L}(t)[1-\Sigma(t)]^{-1}\mathcal{P}} \tag{28}$$

is the **TCL generator**, and

$$\boxed{\mathcal{I}(t)=\alpha\mathcal{P}\mathcal{L}(t)[1-\Sigma(t)]^{-1}\mathcal{G}(t,t_0)\mathcal{Q}} \tag{29}$$

is the **inhomogeneity**.

2.1 Perturbative Expansion of the TCL Generator

Next we derive the perturbative expansion of the TCL master equation. In this note we focus on the cases with factorizing initial condition, where $\mathcal{P}\hat{\rho}(t_0)=\hat{\rho}(t_0)$ and the inhomogeneous term is zero. In this case, we only need to calculate the perturbation expansion of $\mathcal{K}(t)$. Formally,

$$\mathcal{K}(t)\equiv\sum_{n=1}^{\infty}\alpha^n\mathcal{K}_n(t) \tag{30}$$

According to Eq. (28), the leading term in the expansion should be linear in α. Thus the summation starts from $n=1$. We can write $[1-\Sigma(t)]^{-1}$ as geometric series

$$[1-\Sigma(t)]^{-1}=\sum_{n=0}^{\infty}[\Sigma(t)]^n \tag{31}$$

Thus

$$\mathcal{K}(t)=\alpha\sum_{n=0}^{\infty}\mathcal{P}\mathcal{L}(t)[\Sigma(t)]^n\mathcal{P} \tag{32}$$

Further expanding $\mathcal{G}(t,t')$ and $G(t,t')$ as powers of α. For $\mathcal{G}(t,t')$,

$$\mathcal{G}(t,t') = 1 + \alpha \int_{t'}^{t} \mathrm{d}\tau\, \mathcal{QL}(\tau) + \frac{\alpha^2}{2} \hat{T}_{\leftarrow} \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1) + O(\alpha^3) \tag{33}$$

For the second-order term, the time ordering leads to

$$\begin{aligned}
&\hat{T}_{\leftarrow} \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1) \\
=& \int_{\tau_1}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1) + \int_{t'}^{\tau_1} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_1)\mathcal{QL}(\tau_2) \\
=& \int_{\tau_1}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1) + \int_{t'}^{\tau_2} \mathrm{d}\tau_1 \int_{t'}^{t} \mathrm{d}\tau_2\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1) \\
=& 2 \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{\tau_2} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1)
\end{aligned} \tag{34}$$

The last equity can be proved by plotting the integration limits $\int_{\tau_1}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1$ and $\int_{t'}^{\tau_2} \mathrm{d}\tau_1 \int_{t'}^{t} \mathrm{d}\tau_2$ on the $\tau_1\tau_2$ plane. It is evident that the two terms represent the integration in the same area on the $\tau_1\tau_2$ plane. Thus

$$\mathcal{G}(t,t') = 1 + \alpha \int_{t'}^{t} \mathrm{d}\tau\, \mathcal{QL}(\tau) + \alpha^2 \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{\tau_2} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1) + O(\alpha^3) \tag{35}$$

Similarly,

$$G(t,t') = 1 - \alpha \int_{t'}^{t} \mathrm{d}\tau\, \mathcal{L}(\tau) + \alpha^2 \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{\tau_2} \mathrm{d}\tau_1\, \mathcal{L}(\tau_1)\mathcal{L}(\tau_2) + O(\alpha^3) \tag{36}$$

Using the perturbative expansions of $\mathcal{G}(t,t')$ and $G(t,t')$, and the definition of $\Sigma(t)$, we have

$$\begin{aligned}
\Sigma(t) = \alpha &\int_{t_0}^{t} \mathrm{d}t'\, \mathcal{QL}(t')\mathcal{P} \\
&+ \alpha^2 \int_{t_0}^{t} \mathrm{d}t' \int_{t'}^{t} \mathrm{d}\tau\, \left[\mathcal{QL}(\tau)\mathcal{QL}(t')\mathcal{P} - \mathcal{QL}(t')\mathcal{PL}(\tau)\right] \\
&+ \alpha^3 \int_{t_0}^{t} \mathrm{d}t' \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{\tau_2} \mathrm{d}\tau_1\, \left[\mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1)\mathcal{QL}(t')\mathcal{P} + \mathcal{QL}(t')\mathcal{PL}(\tau_1)\mathcal{L}(\tau_2)\right] \\
&- \alpha^3 \int_{t_0}^{t} \mathrm{d}t' \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_2)\mathcal{QL}(t')\mathcal{PL}(\tau_1) \\
&+ O(\alpha^4)
\end{aligned} \tag{37}$$

Next, we rewrite the integration limits so that the lower bound is always t_0. In the second line, we have already discussed that $\int_{t_0}^{t} \mathrm{d}t' \int_{t'}^{t} \mathrm{d}\tau = \int_{t_0}^{t} \mathrm{d}\tau \int_{t_0}^{\tau} \mathrm{d}t'$. In the third line, we have

$$\int_{t_0}^{t} \mathrm{d}t' \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{\tau_2} \mathrm{d}\tau_1 \cdots = \int_{t_0}^{t} \mathrm{d}\tau_2 \int_{t_0}^{\tau_2} \mathrm{d}\tau_1 \int_{t_0}^{\tau_1} \mathrm{d}t' \cdots \tag{38}$$

because both represent integration in the region $t_0 < t' < \tau_1 < \tau_2 < t$. In the fourth line, the integration region is $t_0 < t' < \tau_1 < t$ and $t_0 < t' < \tau_2 < t$, considering the two cases $\tau_1 < \tau_2$ and $\tau_2 < \tau_1$, we have

$$\int_{t_0}^{t} \mathrm{d}t' \int_{t'}^{t} \mathrm{d}\tau_2 \int_{t'}^{t} \mathrm{d}\tau_1 \cdots = \int_{t_0}^{t} \mathrm{d}\tau_2 \int_{t_0}^{\tau_2} \mathrm{d}\tau_1 \int_{t_0}^{\tau_1} \mathrm{d}t' \cdots + \int_{t_0}^{t} \mathrm{d}\tau_1 \int_{t_0}^{\tau_1} \mathrm{d}\tau_2 \int_{t_0}^{\tau_2} \mathrm{d}t' \cdots \tag{39}$$

The final expression for the perturbative expansion of $\Sigma(t)$ is thus

$$\begin{aligned}
\Sigma(t) = \alpha & \int_{t_0}^{t} \mathrm{d}\tau_1\, \mathcal{QL}(\tau_1)\mathcal{P} \\
& + \alpha^2 \int_{t_0}^{t} \mathrm{d}\tau_2 \int_{t_0}^{\tau_2} \mathrm{d}\tau_1\, [\mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1)\mathcal{P} - \mathcal{QL}(\tau_1)\mathcal{PL}(\tau_2)] \\
& + \alpha^3 \int_{t_0}^{t} \mathrm{d}\tau_3 \int_{t_0}^{\tau_3} \mathrm{d}\tau_2 \int_{t_0}^{\tau_2} \mathrm{d}\tau_1\, [\mathcal{QL}(\tau_3)\mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1)\mathcal{P} + \mathcal{QL}(\tau_1)\mathcal{PL}(\tau_2)\mathcal{L}(\tau_3) \\
& \qquad - \mathcal{QL}(\tau_3)\mathcal{QL}(\tau_1)\mathcal{PL}(\tau_2) - \mathcal{QL}(\tau_2)\mathcal{QL}(\tau_1)\mathcal{PL}(\tau_3)] \\
& + O(\alpha^4) \\
\equiv & \ \alpha\Sigma_1(t) + \alpha^2\Sigma_2(t) + \alpha^3\Sigma_3(t) + O(\alpha^4)
\end{aligned} \tag{40}$$

Substituting the expansion of $\Sigma(t)$ into Eq. (32), we obtain the following expressions of $\mathcal{K}_n(t)$ in terms of $\Sigma_n(t)$ (up to the forth-order),

$$\mathcal{K}_1(t) = \mathcal{PL}(t)\mathcal{P} \tag{41a}$$

$$\mathcal{K}_2(t) = \mathcal{PL}(t)\Sigma_1(t)\mathcal{P} \tag{41b}$$

$$\mathcal{K}_3(t) = \mathcal{PL}(t)\{[\Sigma_1(t)]^2 + \Sigma_2(t)\}\mathcal{P} \tag{41c}$$

$$\mathcal{K}_4(t) = \mathcal{PL}(t)\{[\Sigma_1(t)]^3 + \Sigma_1(t)\Sigma_2(t) + \Sigma_2(t)\Sigma_1(t) + \Sigma_3(t)\}\mathcal{P} \tag{41d}$$

The projection operators have the property $\mathcal{PQ} = 0$. Assuming that Eq. (16) holds, then $\mathcal{PL}(t)\mathcal{Q} = \mathcal{PL}(t)$, and $\mathcal{QL}(t)\mathcal{P} = \mathcal{L}(t)\mathcal{P}$. The final results for $\mathcal{K}_n$ are

$$\boxed{\mathcal{K}_1(t) = \mathcal{K}_3(t) = 0} \tag{42a}$$

$$\boxed{\mathcal{K}_2(t) = \int_{t_0}^{t} \mathrm{d}\tau_1\, \mathcal{PL}(t)\mathcal{L}(\tau_1)\mathcal{P}} \tag{42b}$$

$$\boxed{\begin{aligned}\mathcal{K}_4(t) = \int_{t_0}^{t} \mathrm{d}\tau_3 \int_{t_0}^{\tau_3} \mathrm{d}\tau_2 \int_{t_0}^{\tau_2} \mathrm{d}\tau_1\, & [\mathcal{PL}(t)\mathcal{L}(\tau_3)\mathcal{L}(\tau_2)\mathcal{L}(\tau_1)\mathcal{P} - \mathcal{PL}(t)\mathcal{L}(\tau_3)\mathcal{PL}(\tau_2)\mathcal{L}(\tau_1)\mathcal{P} \\ & - \mathcal{PL}(t)\mathcal{L}(\tau_1)\mathcal{PL}(\tau_3)\mathcal{L}(\tau_2)\mathcal{P} - \mathcal{PL}(t)\mathcal{L}(\tau_2)\mathcal{PL}(\tau_3)\mathcal{L}(\tau_1)\mathcal{P}]\end{aligned}} \tag{42c}$$

The general expression of $\mathcal{K}_n(t)$ is

$$\mathcal{K}_n(t) = \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \cdots \int_{t_0}^{t_{n-2}} \mathrm{d}t_{n-1}\, \langle \mathcal{L}(t)\mathcal{L}(t_1)\mathcal{L}(t_2)\cdots\mathcal{L}(t_{n-1})\rangle_{\text{oc}} \tag{43}$$

where

$$\langle \mathcal{L}(t)\mathcal{L}(t_1)\mathcal{L}(t_2)\cdots\mathcal{L}(t_{n-1})\rangle_{\text{oc}} = \sum (-1)^q \mathcal{PL}(t)\cdots\mathcal{L}(t_i)\mathcal{PL}(t_j)\cdots\mathcal{L}(t_k)\mathcal{PL}(t_l)\cdots\mathcal{L}(t_m)\mathcal{P}\cdots\mathcal{P} \tag{44}$$

is the **ordered cumulants**, which can be constructed according to the following rules.

1. Write down a string of the form $\mathcal{PL}\cdots\mathcal{LP}$ with n factors of $\mathcal{L}$ in between two $\mathcal{P}$'s.

2. Insert an arbitrary number q of factors $\mathcal{P}$ between the $\mathcal{L}$'s such that at least one $\mathcal{L}$ stands between two successive $\mathcal{P}$'s, the resulted expression is then multiplied by the factor $(-1)^q$.

3. The first time argument of $\mathcal{L}$ is always t.

4. The remaining $\mathcal{L}$'s carry any permutation of the time arguments $t_1, t_2, \cdots, t_{n-1}$, with the only restriction that the **time arguments between two successive $\mathcal{P}$'s must be ordered chronologically**. In the expression above, for example, we must have $t \geq \cdots \geq t_i$, $t_j \geq \cdots \geq t_k$, $t_l \geq \cdots \geq t_m$, etc.

5. The ordered cumulant is obtained by a summation over all possible insertions of $\mathcal{P}$'s and over all allowed permutation of the time arguments.

Following these rules, it is easy to show that **if the relation Eq. (16) holds, then the odd contributions $\mathcal{K}_{2n+1}(t) = 0$, because any possible insertion of factors $\mathcal{P}$ between the $\mathcal{L}$'s will result in at least one occasion that odd number of $\mathcal{L}$ factors are between two successive $\mathcal{P}$'s**.

3 TCL2 Master Equation in Eigenstate Basis

In this section we focus on the second-order time-convolutionless (TCL2) master equation. We truncate the perturbative expansion of the TCL generator to the second-order, and assume a factorizing initial condition so that the inhomogeneous term vanishes. The equation of motion of $\mathcal{P}\hat{\rho}(t)$ is

$$\boxed{\frac{\partial}{\partial t}\mathcal{P}\hat{\rho}(t) = \alpha^2 \int_{t_0}^{t} \mathrm{d}t'\, \mathcal{P}\mathcal{L}(t)\mathcal{L}(t')\mathcal{P}\hat{\rho}(t)} \tag{45}$$

or explicitly in terms of the commutators

$$\boxed{\frac{\partial}{\partial t}\hat{\rho}_{\mathrm{S}}(t) = -\alpha^2 \int_{t_0}^{t} \mathrm{d}t'\, \mathrm{Tr}_{\mathrm{B}}\left([\hat{H}_{\mathrm{I}}(t), [\hat{H}_{\mathrm{I}}(t'), \hat{\rho}_{\mathrm{S}}(t) \otimes \hat{\rho}_{\mathrm{B}}]]\right)} \tag{46}$$

Note that the time argument for $\hat{\rho}_{\mathrm{S}}$ is t in this equation, in contrast to t' in Eq. (20). The pre-factor α^2 signifies that this equation it to the second-order of the system–bath coupling. In the subsequent derivations, we set $\alpha = 1$.

Assuming that the system–bath coupling has the form

$$\hat{H}_{\mathrm{I}} = \sum_m \hat{S}_m \otimes \hat{B}_m \tag{47}$$

where $\hat{S}_m$ is an operator that only acts on the system and $\hat{B}_m$ is an operator that only acts on the bath. $\hat{S}_m$ and $\hat{B}_m$ are not necessarily Hermitian. $\hat{S}_m(t) = e^{i\hat{H}_{\mathrm{S}}t}\hat{S}_m e^{-i\hat{H}_{\mathrm{S}}t}$ and $\hat{B}_m(t) = e^{i\hat{H}_{\mathrm{B}}t}\hat{B}_m e^{-i\hat{H}_{\mathrm{B}}t}$ are these operators in the interaction picture. Let $\hat{\rho}_{\mathrm{B}} = \frac{\exp(-\beta\hat{H}_{\mathrm{B}})}{\mathrm{Tr}_{\mathrm{B}}\exp(-\beta\hat{H}_{\mathrm{B}})}$ be the equilibrium density operator of the bath. The bath correlation function is defined as

$$C_{mn}(t) = \mathrm{Tr}_{\mathrm{B}}\left(\hat{B}_m(t)\hat{B}_n(0)\hat{\rho}_{\mathrm{B}}\right) \tag{48}$$

Substituting this form of $\hat{H}_\mathrm{I}$ and the definition of $C_{mn}(t)$ into the TCL2 master equation, we have (setting $t_0 = 0$)

$$\begin{aligned}
\frac{\partial}{\partial t}\hat{\rho}_\mathrm{S}(t) &= -\sum_{m,n}\int_0^t \mathrm{d}t' \, \mathrm{Tr}_\mathrm{B}\left([\hat{S}_m(t)\otimes\hat{B}_m(t), [\hat{S}_n(t')\otimes\hat{B}_n(t'), \hat{\rho}_\mathrm{S}(t)\otimes\hat{\rho}_\mathrm{B}]]\right) \\
&= -\sum_{m,n}\int_0^t \mathrm{d}t' \, \left\{C_{mn}(t-t')[\hat{S}_m(t), \hat{S}_n(t')\hat{\rho}_\mathrm{S}(t)] - C_{nm}(t'-t)[\hat{S}_m(t), \hat{\rho}_\mathrm{S}(t)\hat{S}_n(t')]\right\} \\
&= -\sum_{m,n}\int_0^t \mathrm{d}\tau \, \left\{C_{mn}(\tau)[\hat{S}_m(t), \hat{S}_n(t-\tau)\hat{\rho}_\mathrm{S}(t)] - C_{nm}(-\tau)[\hat{S}_m(t), \hat{\rho}_\mathrm{S}(t)\hat{S}_n(t-\tau)]\right\}
\end{aligned} \tag{49}$$

Let $|a\rangle$, $|b\rangle$ be the eigenstates of the system Hamiltonian $\hat{H}_\mathrm{S}$, and E_a, E_b are the energies of the eigenstates. Defining $\omega_{ab} = E_a - E_b$, then the TCL2 master equation for the density matrix in the eigenstate basis becomes

$$\begin{aligned}
\frac{\partial}{\partial t}\rho_{ab}(t) = & -i\omega_{ab}\rho_{ab}(t) \\
& -\sum_{m,n}\sum_{c,d}\int_0^t \mathrm{d}\tau \, C_{mn}(\tau)\left[S_{ac}^{(m)}S_{cd}^{(n)}e^{i\omega_{dc}\tau}\rho_{db}(t) - S_{db}^{(m)}S_{ac}^{(n)}e^{i\omega_{ca}\tau}\rho_{cd}(t)\right] \\
& +\sum_{m,n}\sum_{c,d}\int_0^t \mathrm{d}\tau \, C_{nm}(-\tau)\left[S_{ac}^{(m)}S_{db}^{(n)}e^{i\omega_{bd}\tau}\rho_{cd}(t) - S_{db}^{(m)}S_{cd}^{(n)}e^{i\omega_{dc}\tau}\rho_{ac}(t)\right]
\end{aligned} \tag{50}$$

where $S_{ab}^{(m)} = \langle a|\hat{S}_m|b\rangle$. To simplify the equation, we introduce the integrated bath correlation function,

$$\boxed{F_{mn}^{(\pm)}(\omega, t) = \int_0^t \mathrm{d}\tau \, C_{mn}(\pm\tau)e^{i\omega\tau}} \tag{51}$$

If $\hat{B}_m$ is Hermitian, then $C_{mn}(\tau) = C_{nm}^*(-\tau)$, and $\left[F_{nm}^{(-)}(\omega, t)\right]^* = F_{mn}^{(+)}(-\omega, t)$. In general, however, this relation does not hold. We further define the time-dependent damping matrix,

$$\boxed{\Gamma_{abcd}^{(+)}(t) = \sum_{m,n} S_{ab}^{(m)}S_{cd}^{(n)}F_{mn}^{(+)}(\omega_{dc}, t)} \tag{52a}$$

$$\boxed{\Gamma_{abcd}^{(-)}(t) = \sum_{m,n} S_{ab}^{(m)}S_{cd}^{(n)}F_{mn}^{(-)}(\omega_{ba}, t)} \tag{52b}$$

Similarly, if $\hat{S}_m$ and $\hat{B}_m$ are Hermitian, then $\left[S_{ab}^{(m)}\right]^* = S_{ba}^{(m)}$, and thus $\left[\Gamma_{abcd}^{(-)}(t)\right]^* = \Gamma_{dcba}^{(+)}(t)$. Due to the summation over all m and n, **this relation also holds when $\hat{S}_m$ and $\hat{B}_m$ are not Hermitian, as long as $\hat{S}_m^\dagger = \hat{S}_{m'} \in \{\hat{S}_m\}$ and $\hat{B}_m^\dagger = \hat{B}_{m'} \in \{\hat{B}_m\}$ with a one-to-one correspondance between m and m'.**

Using the definitions of the damping matrix, the TCL2 master equation can be simplified as

$$\frac{\partial}{\partial t}\rho_{ab}(t) = -i\omega_{ab}\rho_{ab}(t) - \sum_{c,d}\left[\Gamma_{accd}^{(+)}(t)\rho_{db}(t) - \Gamma_{dbac}^{(+)}(t)\rho_{cd}(t) - \Gamma_{dbac}^{(-)}(t)\rho_{cd}(t) + \Gamma_{cddb}^{(-)}(t)\rho_{ac}(t)\right] \tag{53}$$

By changing the dummy indices in the summation, we can express the second term on the right hand side of the equation in terms of $\rho_{cd}(t)$. The final results is

$$\boxed{\frac{\partial}{\partial t}\rho_{ab}(t) = -i\omega_{ab}\rho_{ab}(t) + \sum_{c,d}\mathcal{R}_{abcd}(t)\rho_{cd}(t)} \tag{54}$$

where

$$\boxed{\mathcal{R}_{abcd}(t) = \Gamma^{(+)}_{dbac}(t) + \Gamma^{(-)}_{dbac}(t) - \delta_{bd}\sum_{e}\Gamma^{(+)}_{aeec}(t) - \delta_{ac}\sum_{e}\Gamma^{(-)}_{deeb}(t)} \tag{55}$$

is the relaxation superoperator. If $\hat{S}_m$ and $\hat{B}_m$ are Hermitian or they satisfy the condition stated above, $\mathcal{R}_{abcd}(t)$ can be expressed as (defining $\Gamma_{abcd} \equiv \Gamma^{(+)}_{abcd}$ for notational simplicity)

$$\mathcal{R}_{abcd}(t) = \Gamma_{dbac}(t) + \Gamma^{*}_{cabd}(t) - \delta_{bd}\sum_{e}\Gamma_{aeec}(t) - \delta_{ac}\sum_{e}\Gamma^{*}_{beed}(t) \tag{56}$$

It is also obvious that the Redfield tensor equals to $\lim_{t\to\infty}\mathcal{R}_{abcd}(t)$.